

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

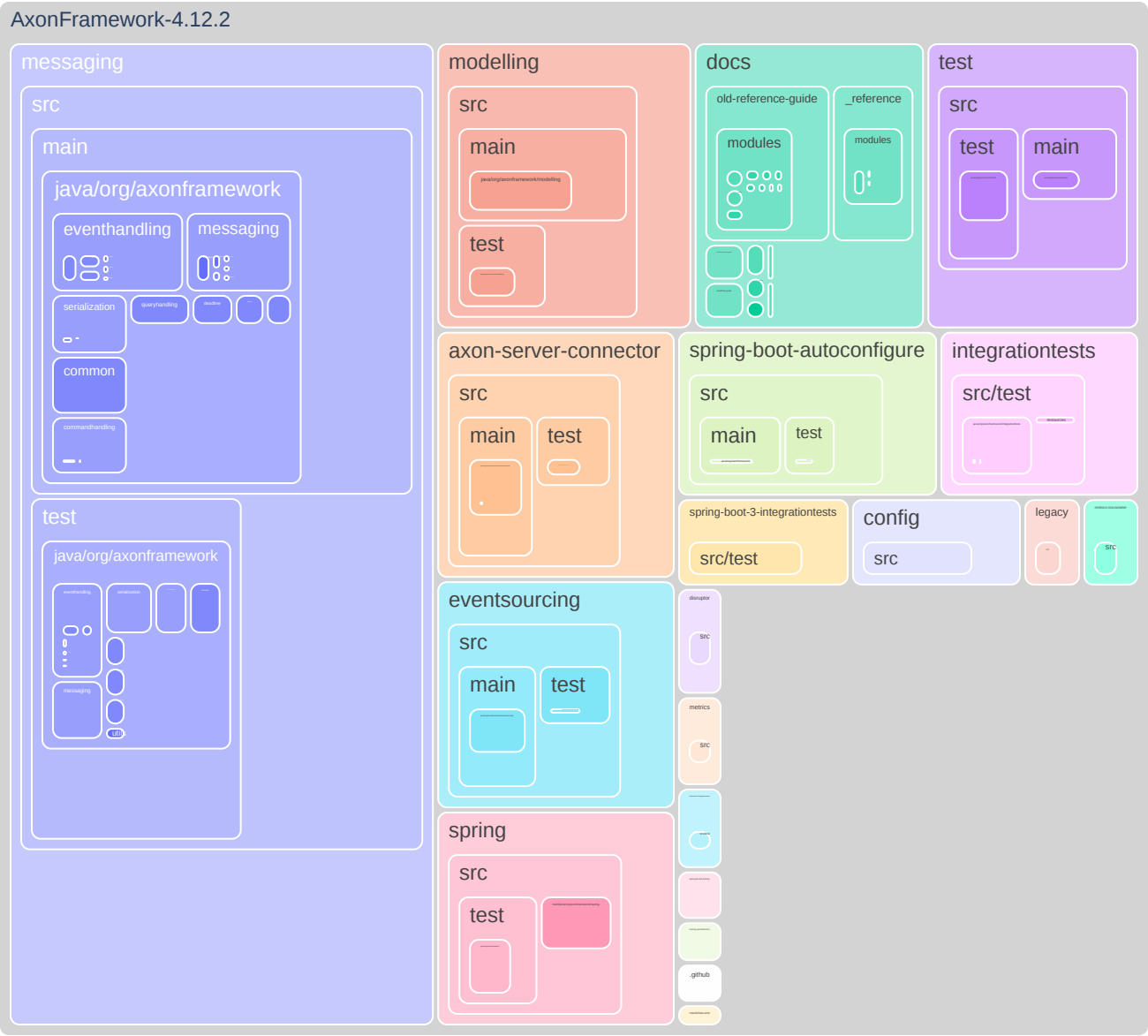
### Data Preview

|       | fileCount   | authorCount | commitCount | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|-------------|-------------|-------------|---------------------|-----------------------|---------------------------|
| count | 566.000000  | 566.000000  | 566.000000  | 566.000000          | 566.000000            | 566.000000                |
| mean  | 24.867491   | 19.606007   | 288.137809  | 366.904594          | 1250.793286           | 460.316254                |
| std   | 120.564946  | 21.286048   | 640.533206  | 187.648018          | 982.294306            | 402.713944                |
| min   | 1.000000    | 2.000000    | 3.000000    | 11.000000           | 15.000000             | 10.000000                 |
| 25%   | 1.250000    | 7.000000    | 27.000000   | 241.000000          | 515.000000            | 302.000000                |
| 50%   | 4.000000    | 12.000000   | 81.500000   | 474.000000          | 1034.000000           | 473.000000                |
| 75%   | 10.000000   | 25.000000   | 269.250000  | 474.000000          | 2050.500000           | 473.000000                |
| max   | 2295.000000 | 213.000000  | 9840.000000 | 2624.000000         | 5752.000000           | 3103.000000               |

|    | directoryPath                                                                                                    | directoryParentPath                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 0  | AxonFramework-4.12.2/assembly                                                                                    | AxonFramework-4.12.                                                                                       |
| 1  | AxonFramework-4.12.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source | AxonFramework-4.12.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat |
| 2  | AxonFramework-4.12.2/axon-server-connector/src/main/resources/META-INF/services                                  | AxonFramework-4.12.2/axon-server-connector/src/main/resources                                             |
| 3  | AxonFramework-4.12.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util       | AxonFramework-4.12.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event     |
| 4  | AxonFramework-4.12.2/config/src/test/resources/META-INF                                                          | AxonFramework-4.12.2/config/src/test/resources                                                            |
| 5  | AxonFramework-4.12.2/coverage-report                                                                             | AxonFramework-4.12.                                                                                       |
| 6  | AxonFramework-4.12.2/docs/_playbook/localLinks                                                                   | AxonFramework-4.12.2/docs/_playbook                                                                       |
| 7  | AxonFramework-4.12.2/docs/_reference/modules/ROOT/attachments                                                    | AxonFramework-4.12.2/docs/_reference/modules/ROOT                                                         |
| 8  | AxonFramework-4.12.2/docs/_reference/modules/ROOT/partials                                                       | AxonFramework-4.12.2/docs/_reference/modules/ROOT                                                         |
| 9  | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector/pages                                         | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector                                        |
| 10 | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector/partials                                      | AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector                                        |
| 11 | AxonFramework-4.12.2/docs/_reference/modules/configuration/pages                                                 | AxonFramework-4.12.2/docs/_reference/modules/configuration                                                |
| 12 | AxonFramework-4.12.2/docs/_reference/modules/configuration/partials                                              | AxonFramework-4.12.2/docs/_reference/modules/configuration                                                |
| 13 | AxonFramework-4.12.2/docs/_reference/modules/disruptor/pages                                                     | AxonFramework-4.12.2/docs/_reference/modules/disruptor                                                    |
| 14 | AxonFramework-4.12.2/docs/_reference/modules/disruptor/partials                                                  | AxonFramework-4.12.2/docs/_reference/modules/disruptor                                                    |
| 15 | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing/pages                                                 | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing                                                |
| 16 | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing/partials                                              | AxonFramework-4.12.2/docs/_reference/modules/eventsourcing                                                |
| 17 | AxonFramework-4.12.2/docs/_reference/modules/legacy/pages                                                        | AxonFramework-4.12.2/docs/_reference/modules/legacy                                                       |
| 18 | AxonFramework-4.12.2/docs/_reference/modules/legacy/partials                                                     | AxonFramework-4.12.2/docs/_reference/modules/legacy                                                       |
| 19 | AxonFramework-4.12.2/docs/_reference/modules/metrics/pages                                                       | AxonFramework-4.12.2/docs/_reference/modules/metrics                                                      |
| 20 | AxonFramework-4.12.2/docs/_reference/modules/metrics/partials                                                    | AxonFramework-4.12.2/docs/_reference/modules/metrics                                                      |
| 21 | AxonFramework-4.12.2/docs/_reference/modules/micrometer/pages                                                    | AxonFramework-4.12.2/docs/_reference/modules/micrometer                                                   |
| 22 | AxonFramework-4.12.2/docs/_reference/modules/micrometer/partials                                                 | AxonFramework-4.12.2/docs/_reference/modules/micrometer                                                   |
| 23 | AxonFramework-4.12.2/docs/_reference/modules/modeling/partials                                                   | AxonFramework-4.12.2/docs/_reference/modules/modeling                                                     |
| 24 | AxonFramework-4.12.2/docs/_reference/modules/spring/pages                                                        | AxonFramework-4.12.2/docs/_reference/modules/spring                                                       |
| 25 | AxonFramework-4.12.2/docs/_reference/modules/spring/partials                                                     | AxonFramework-4.12.2/docs/_reference/modules/spring                                                       |
| 26 | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter/pages                                            | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter                                           |
| 27 | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter/partials                                         | AxonFramework-4.12.2/docs/_reference/modules/springboot-starter                                           |
| 28 | AxonFramework-4.12.2/docs/_reference/modules/test/pages                                                          | AxonFramework-4.12.2/docs/_reference/modules/test                                                         |
| 29 | AxonFramework-4.12.2/docs/_reference/modules/test/partials                                                       | AxonFramework-4.12.2/docs/_reference/modules/test                                                         |

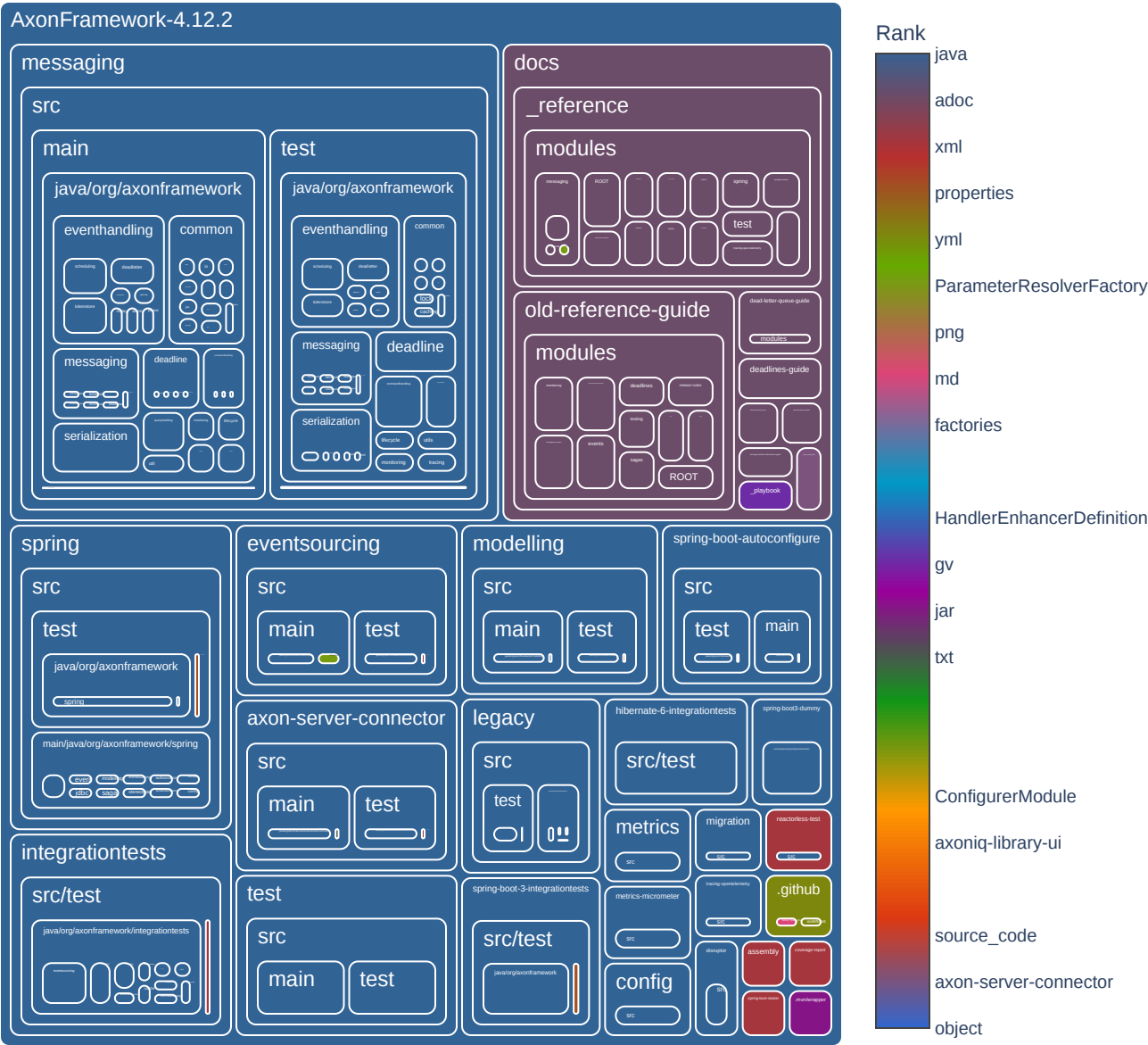
## Number of files per directory

Directories and their file count



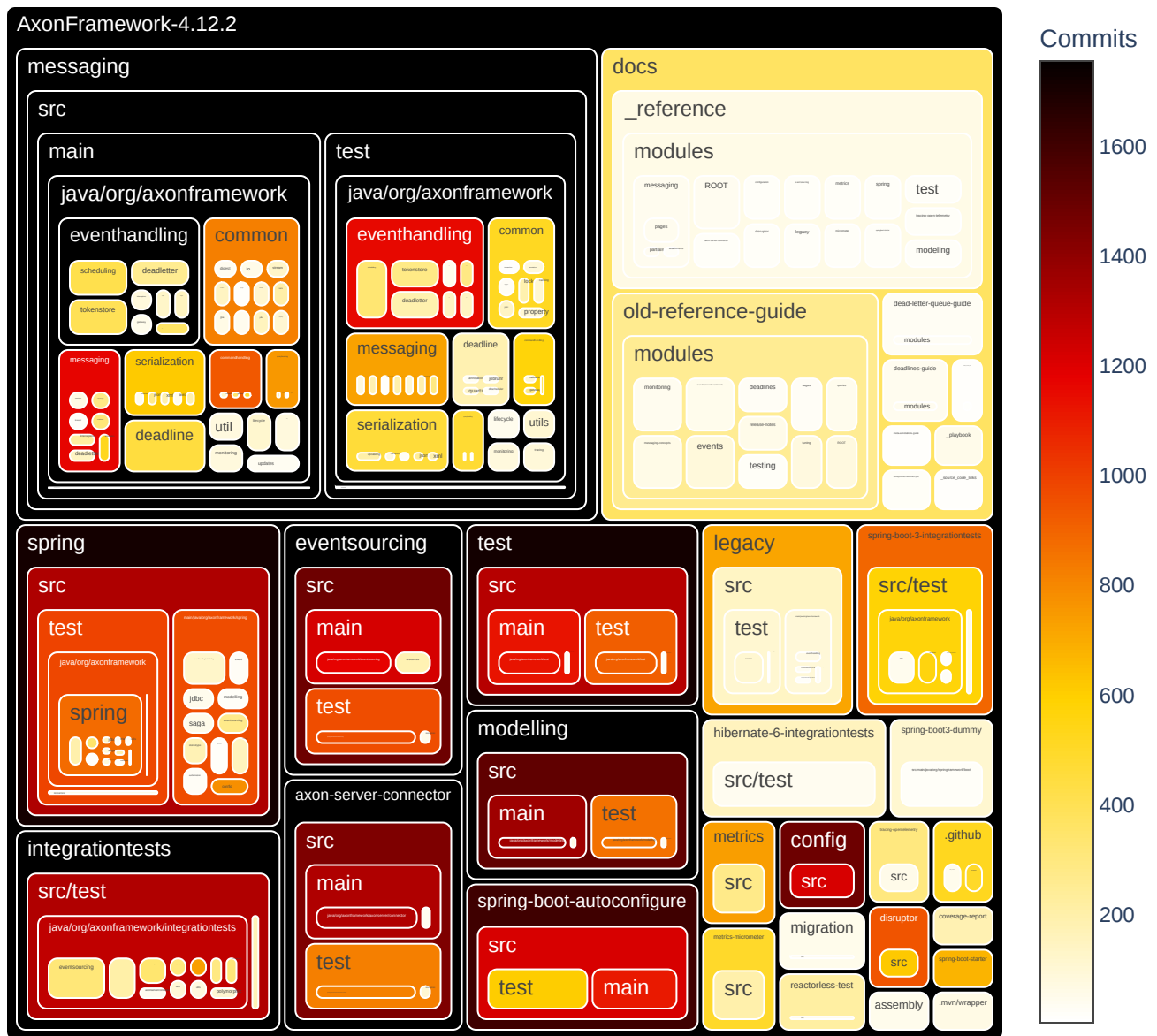
Most frequent file extension per directory

Most frequent file extension per directory



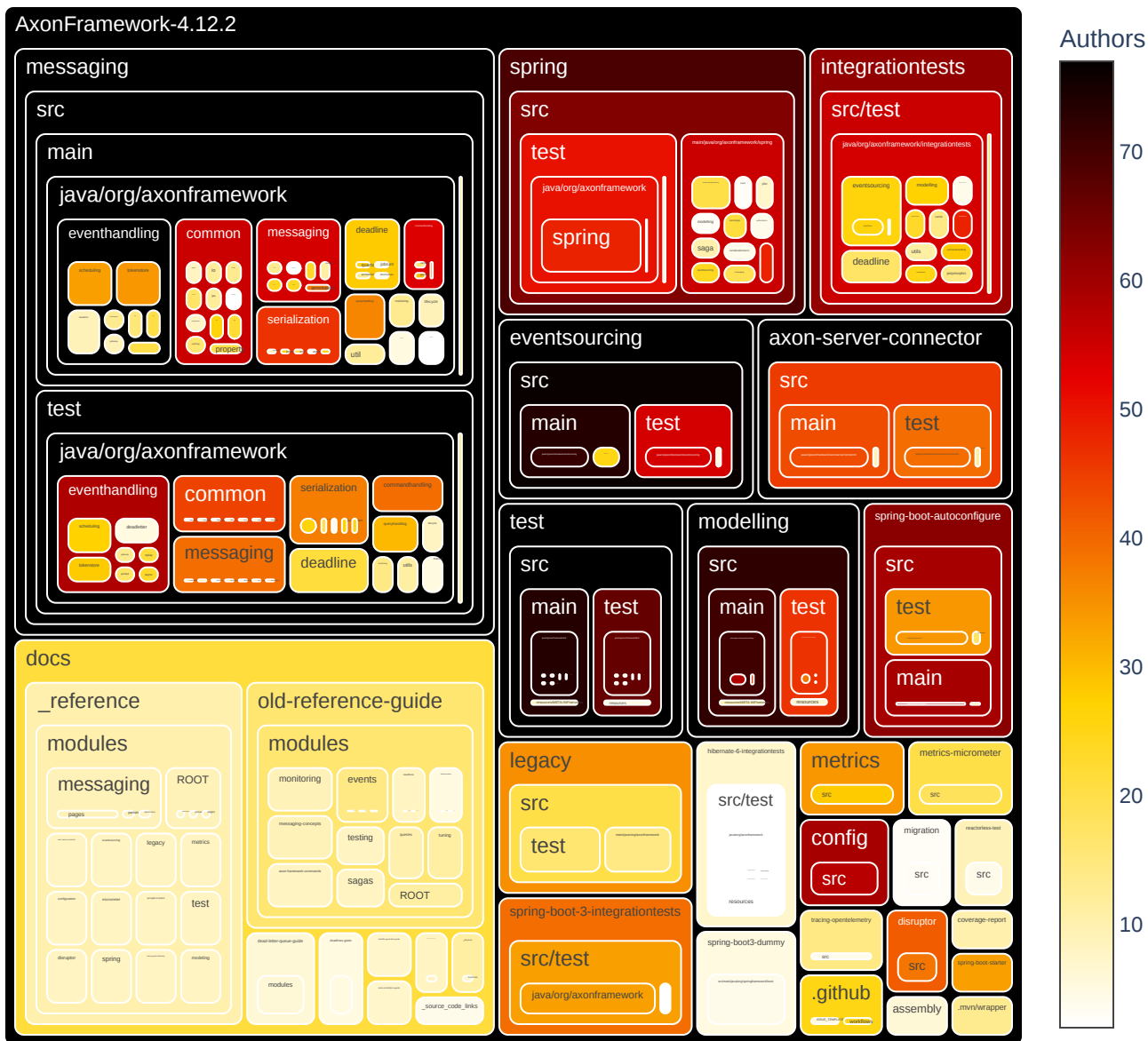
Number of commits per directory

## Number of git commits



## Number of distinct authors per directory

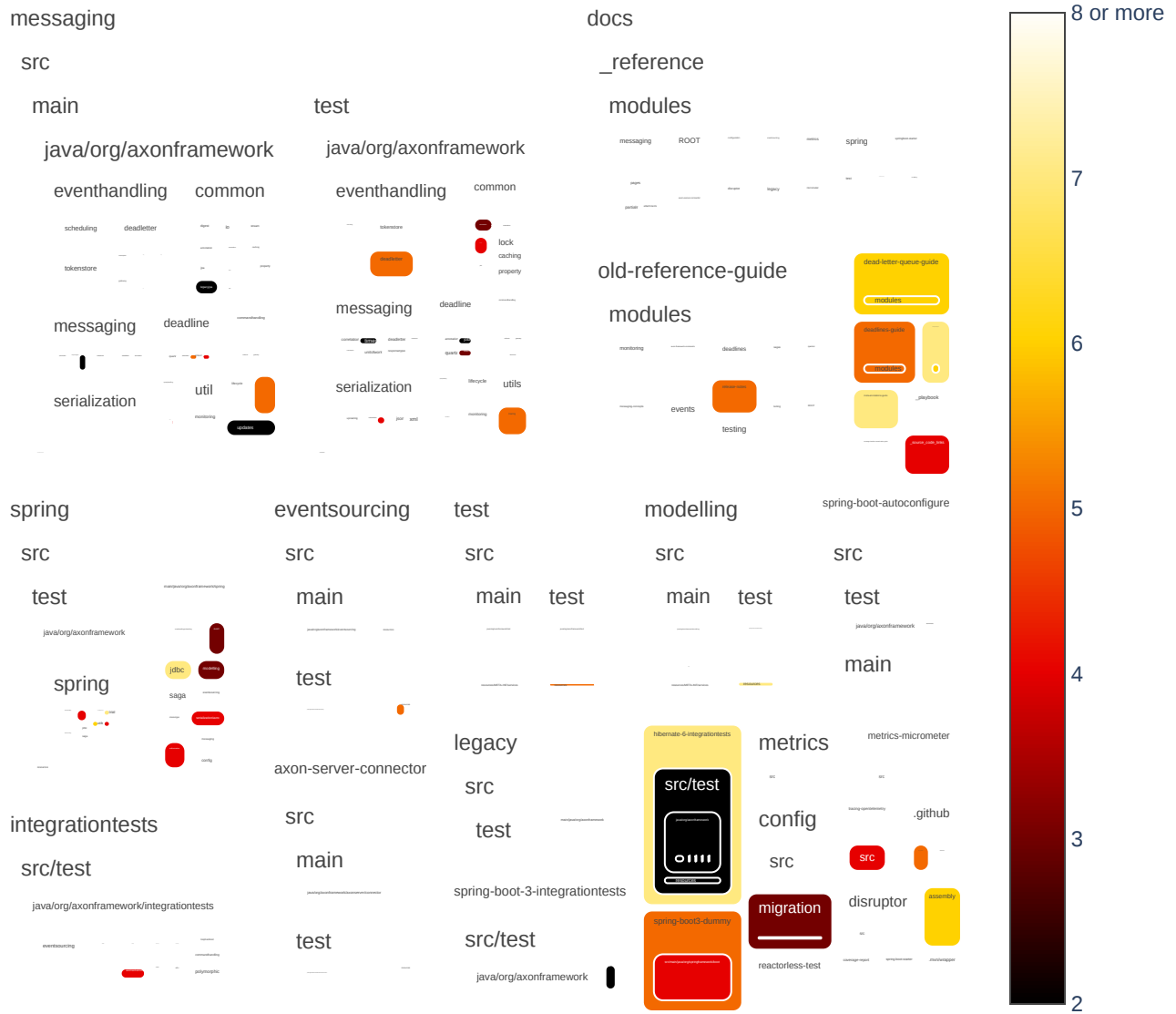
## Number of distinct commit authors



Directories with very few different authors

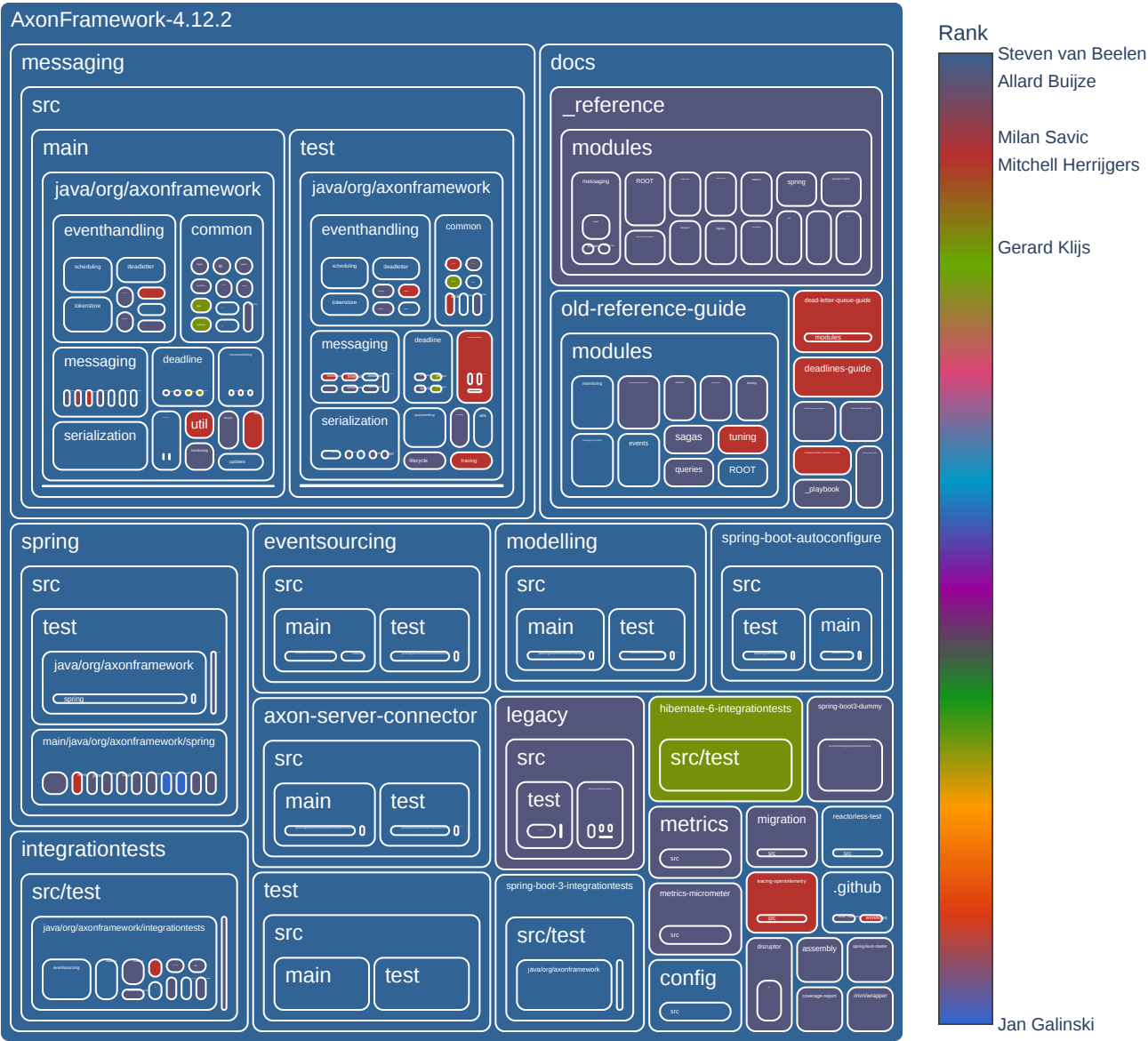
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.12.2



## Main author per directory

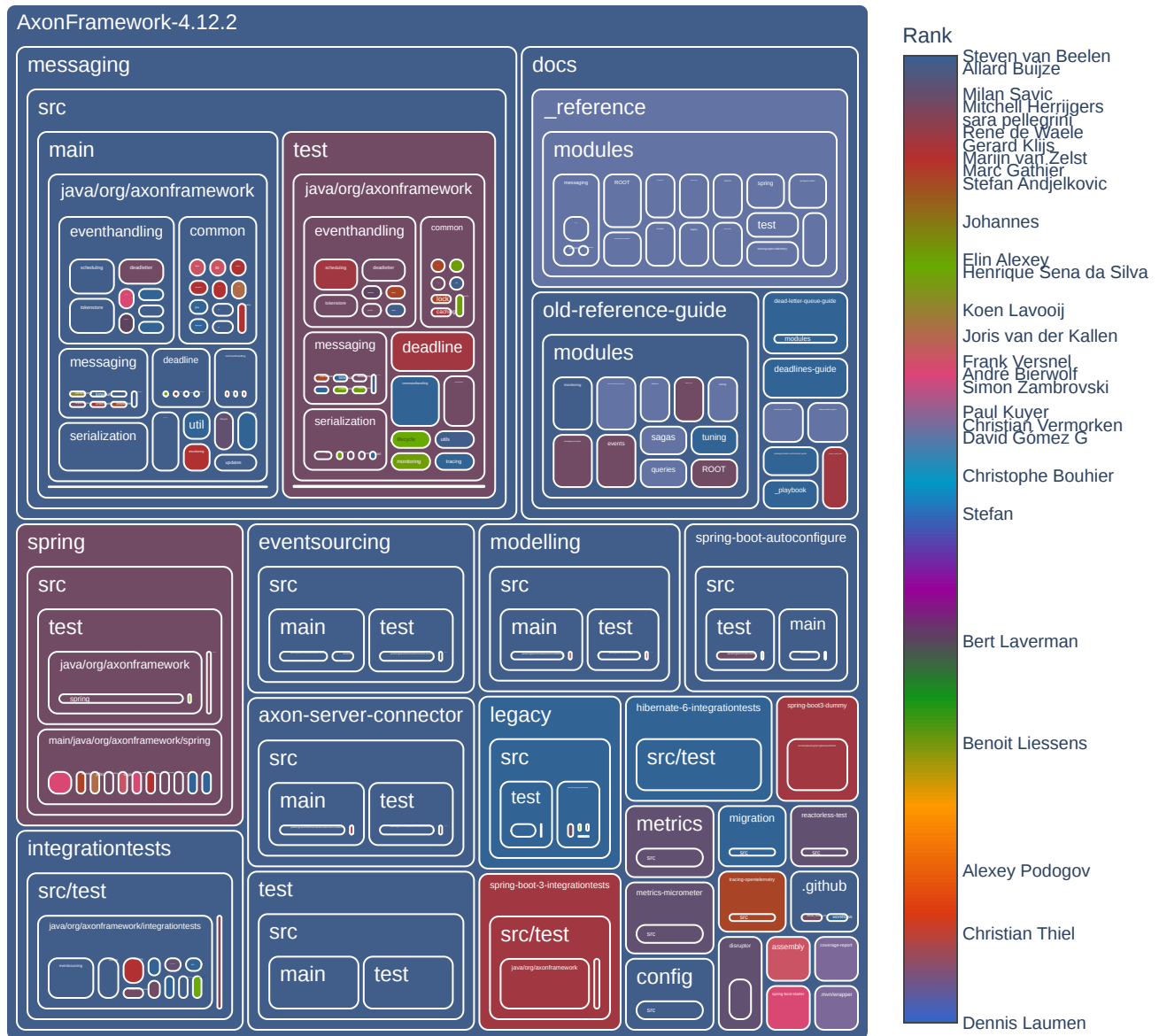
Main authors with highest number of commits



Second author per directory



### Second author with the second highest number of commits



## Days since last commit per directory

Days since last commit

AxonFramework-4.12.2

messaging

src

main

java/org/axonframework

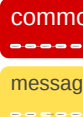
eventhandling



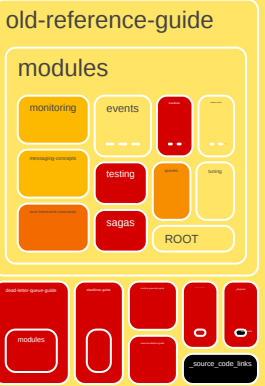
test

java/org/axonframework

eventhandling



docs



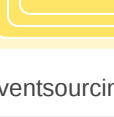
spring

src

test

java/org/axonframework

spring

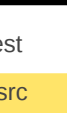


eventsourcing

src

main

test



integrationtests

src/test

java/org/axonframework/integrationtests

eventsourcing

messaging



axon-server-connector

src

main

test



test

src

main

test



modelling

src

main

test



spring-boot-autoconfigure

src

test

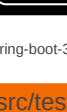
main



legacy

src

test



spring-boot-3-integrationtests

src/test

java/org/axonframework



hibernate-5-integrationtests

src/test

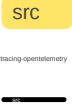
java/org/axonframework



metrics

src

config



metrics-micrometer

src

migration



spring-boot-dummy

src



tracing-opentracing

src



.github

src

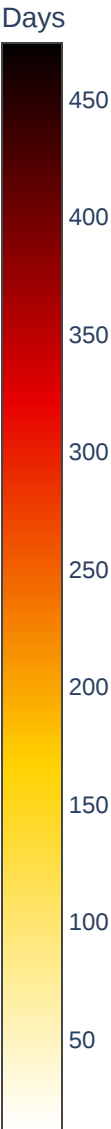


disruptor

src



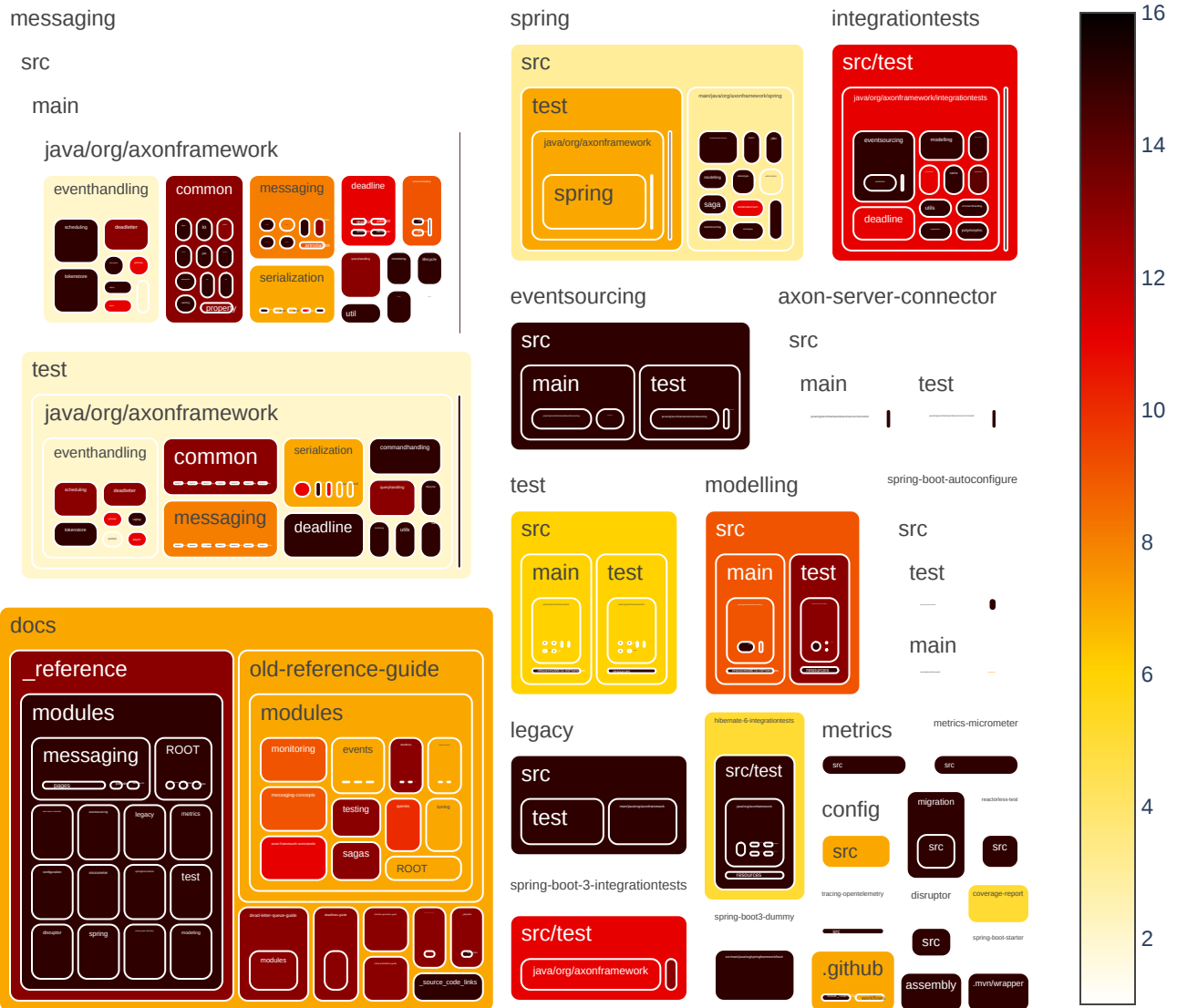
assembly



Days since last commit per directory (ranked)

## Rank of days since last commit

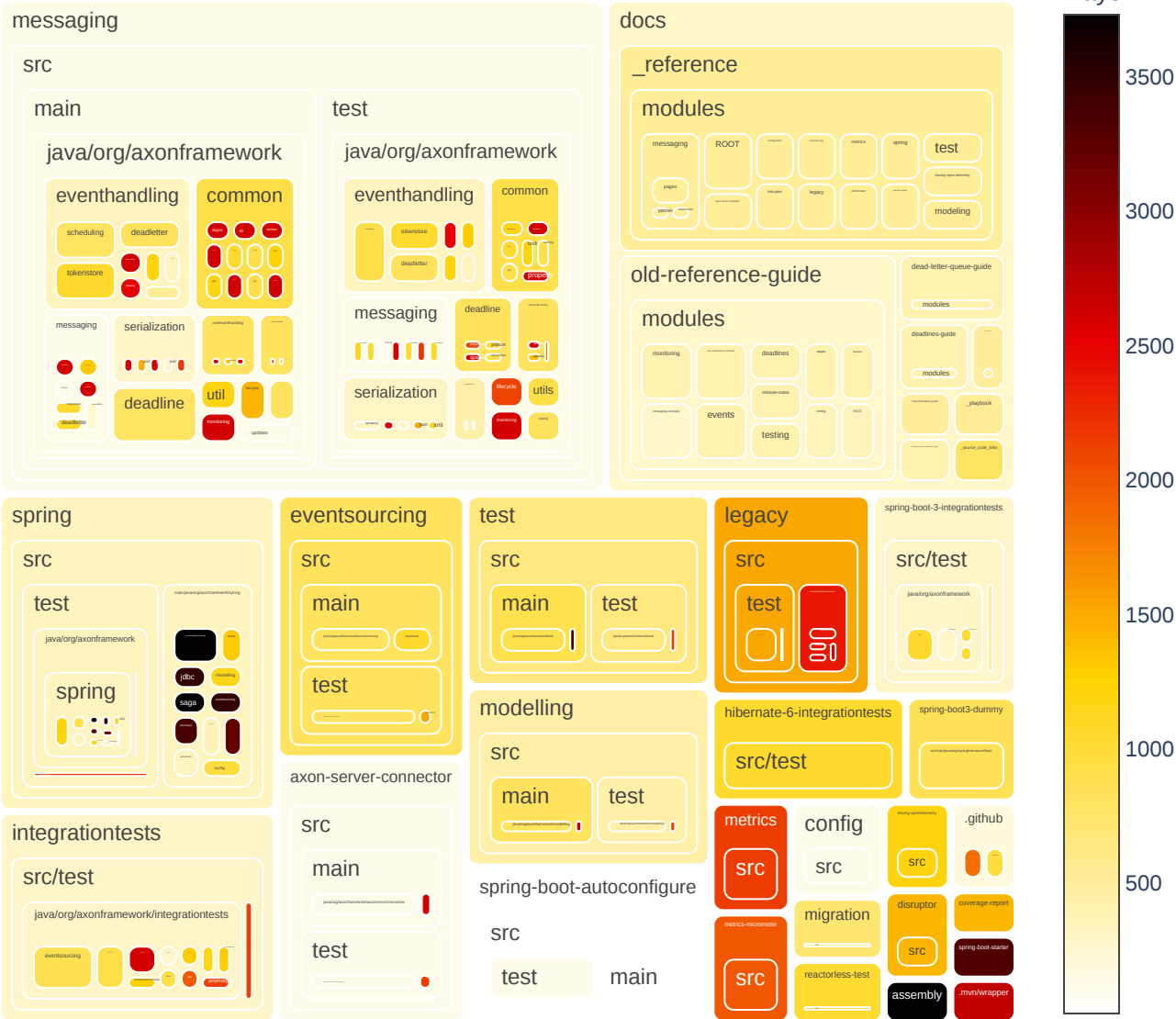
AxonFramework-4.12.2



## Days since last file creation per directory

Days since last file creation

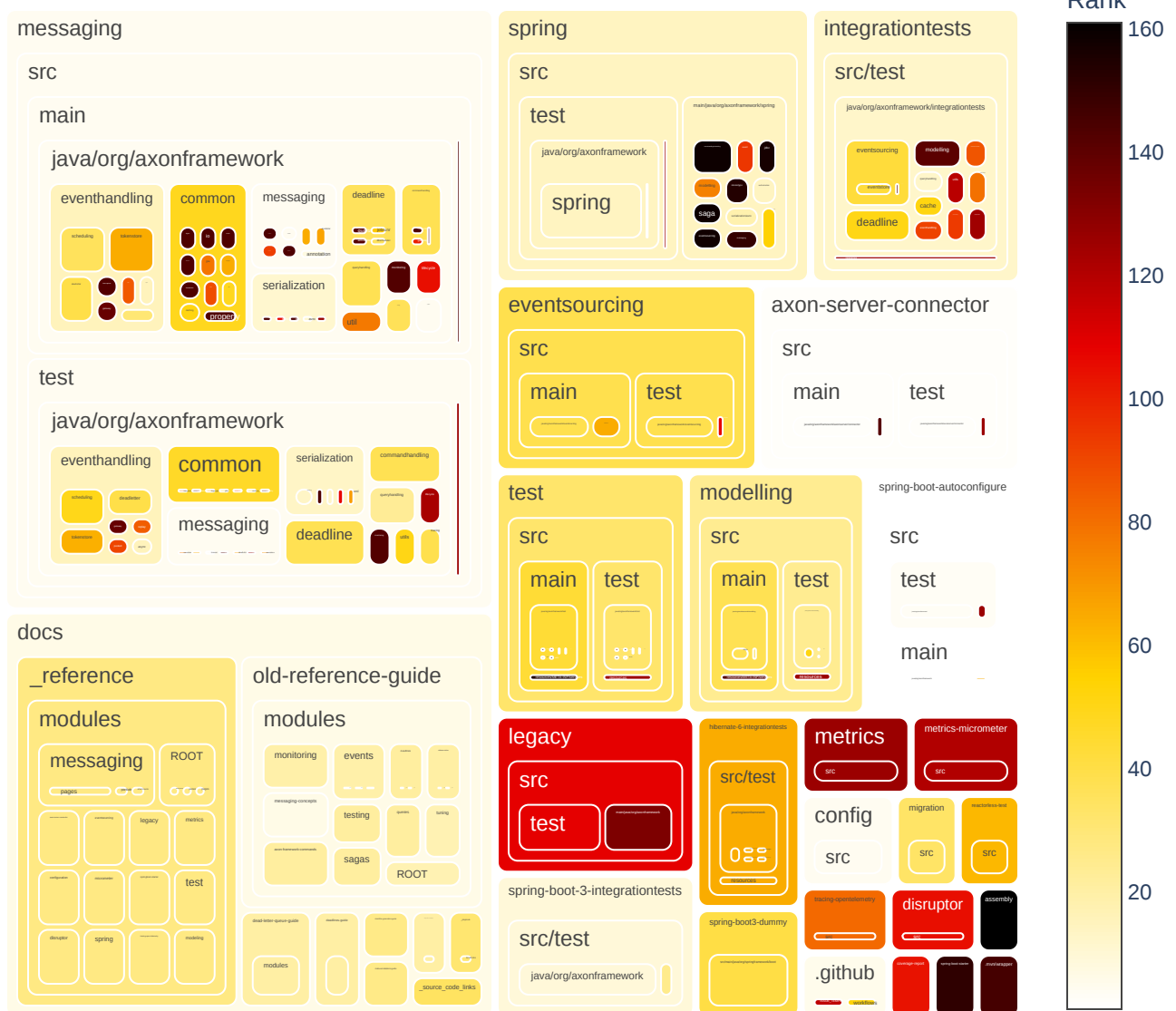
AxonFramework-4.12.2



Days since last file creation per directory (ranked)

## Rank of days since last file creation

AxonFramework-4.12.2



## Days since last file modification per directory

Days since last file modification

AxonFramework-4.12.2

messaging

src

main

java/org/axonframework



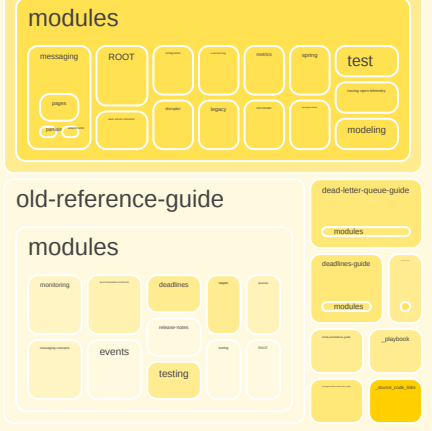
test

java/org/axonframework



docs

\_reference



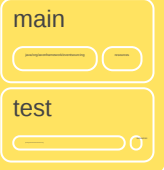
spring

src



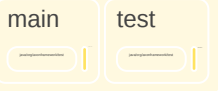
eventsourcing

src



test

src



legacy

src



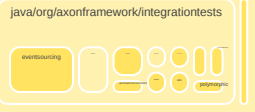
spring-boot-3-integrationtests

src/test



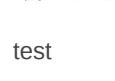
integrationtests

src/test



src

main

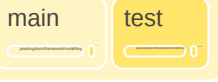


test



modelling

src



spring-boot-autoconfigure

src



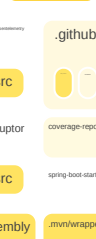
hibernate-6-integrationtests

src/test



spring-boot3-dummy

src



Days



2000

1500

1000

500

Days since last file modification per directory (ranked)

## Rank of days since last file modification

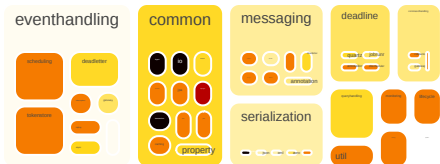
AxonFramework-4.12.2

messaging

src

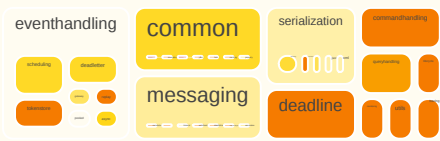
main

java/org/axonframework



test

java/org/axonframework



docs

\_reference



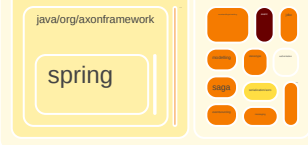
old-reference-guide



spring

src

test



integrationtests

src/test



eventsourcing

src



axon-server-connector

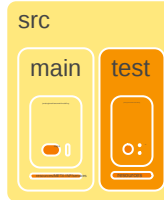
src



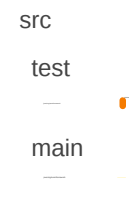
test



modelling



spring-boot-autoconfigure



legacy



spring-boot-3-integrationtests



hibernate-6-integrationtests



metrics



metrics-micrometer



Rank

50

40

30

20

10

## Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

## Preview data

Sum of commits that changed more than 30 files (each) = 973

Max changed files with one commit = 2178

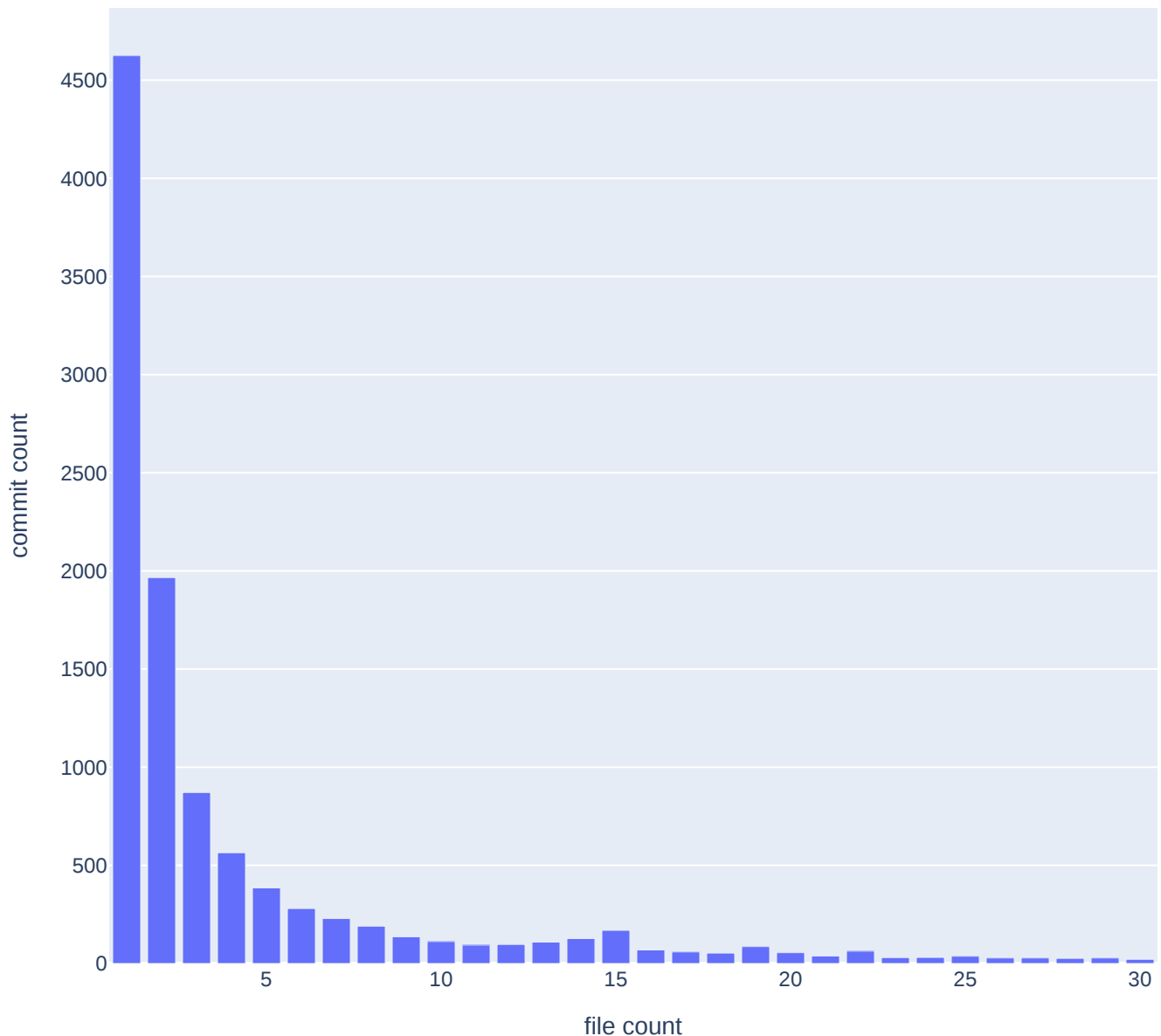
|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 334.000000     | 334.000000  |
| mean  | 343.449102     | 34.667665   |
| std   | 439.862566     | 281.534044  |
| min   | 1.000000       | 1.000000    |
| 25%   | 84.250000      | 1.000000    |
| 50%   | 185.000000     | 2.000000    |
| 75%   | 426.500000     | 5.000000    |
| max   | 2178.000000    | 4625.000000 |

|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 4625        |
| 1  | 2              | 1966        |
| 2  | 3              | 871         |
| 3  | 4              | 564         |
| 4  | 5              | 385         |
| 5  | 6              | 280         |
| 6  | 7              | 229         |
| 7  | 8              | 190         |
| 8  | 9              | 136         |
| 9  | 10             | 112         |
| 10 | 11             | 94          |
| 11 | 12             | 97          |
| 12 | 13             | 109         |
| 13 | 14             | 127         |
| 14 | 15             | 169         |
| 15 | 16             | 69          |
| 16 | 17             | 59          |
| 17 | 18             | 53          |
| 18 | 19             | 86          |
| 19 | 20             | 55          |
| 20 | 21             | 38          |
| 21 | 22             | 63          |
| 22 | 23             | 30          |
| 23 | 24             | 31          |
| 24 | 25             | 37          |
| 25 | 26             | 28          |
| 26 | 27             | 28          |
| 27 | 28             | 26          |
| 28 | 29             | 28          |
| 29 | 30             | 21          |

Bar chart with the number of files per commit distribution



## Changed files per commit



## Pairwise Changed Files

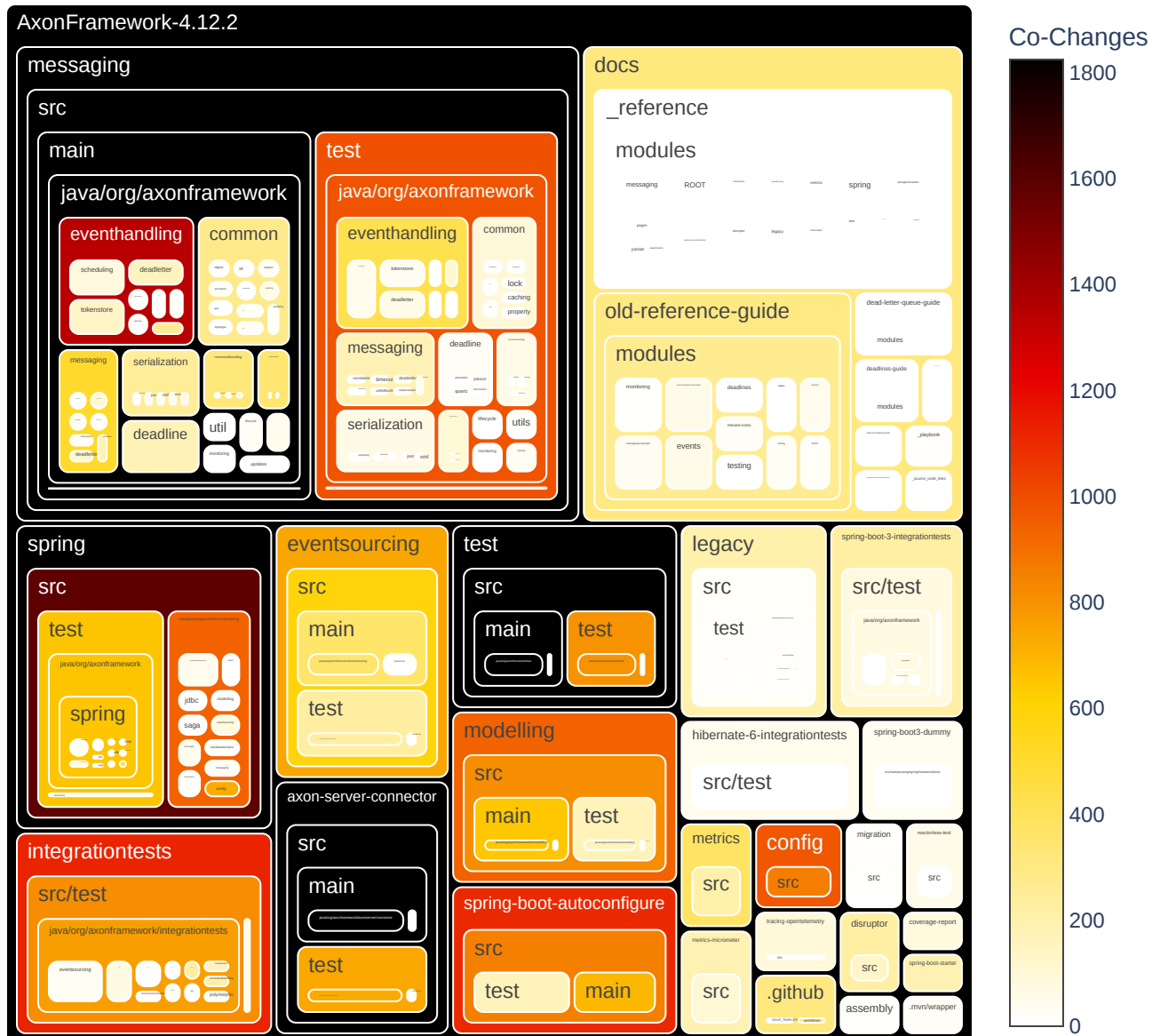
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

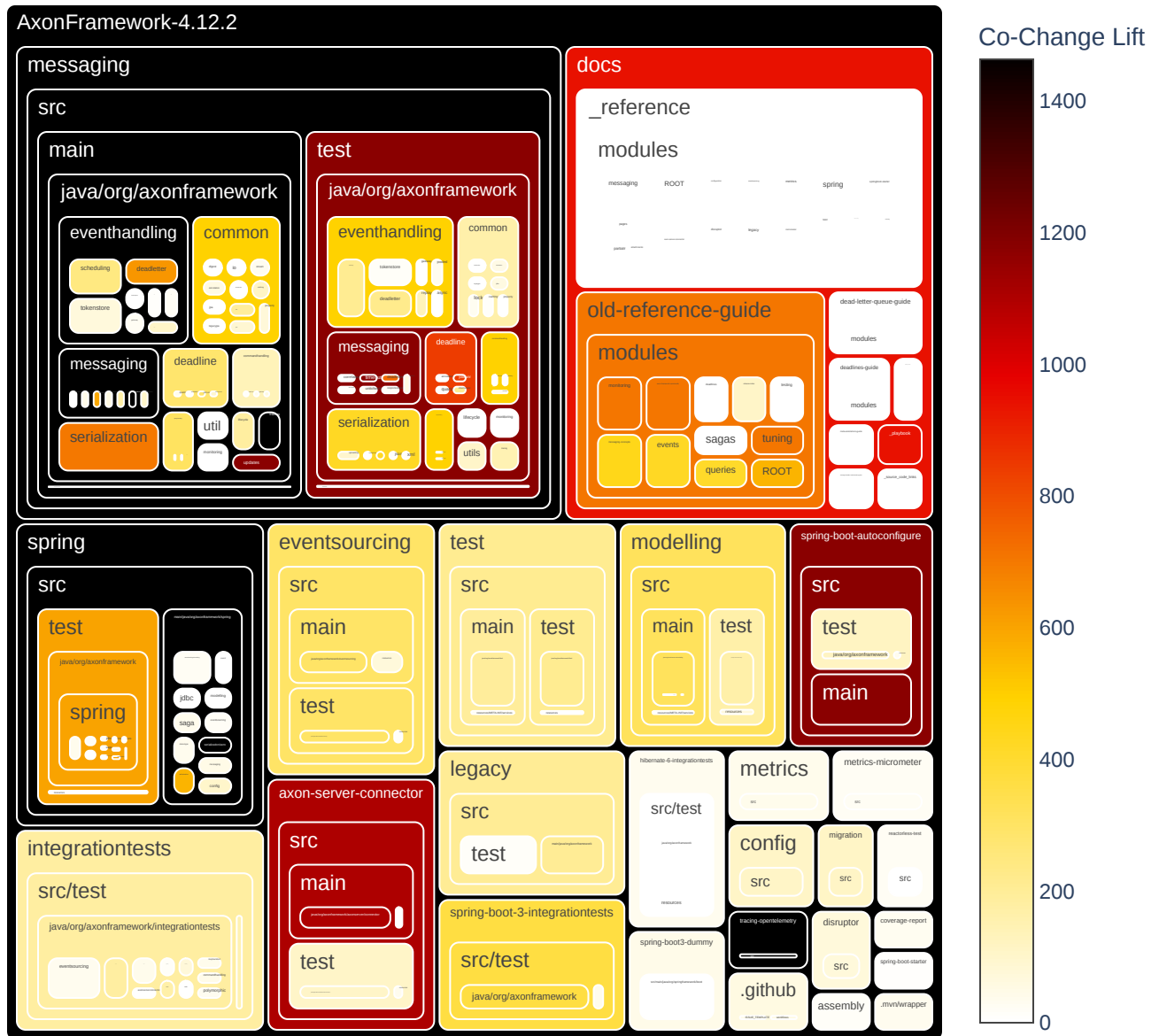
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

## Treemap with files changed frequently with others

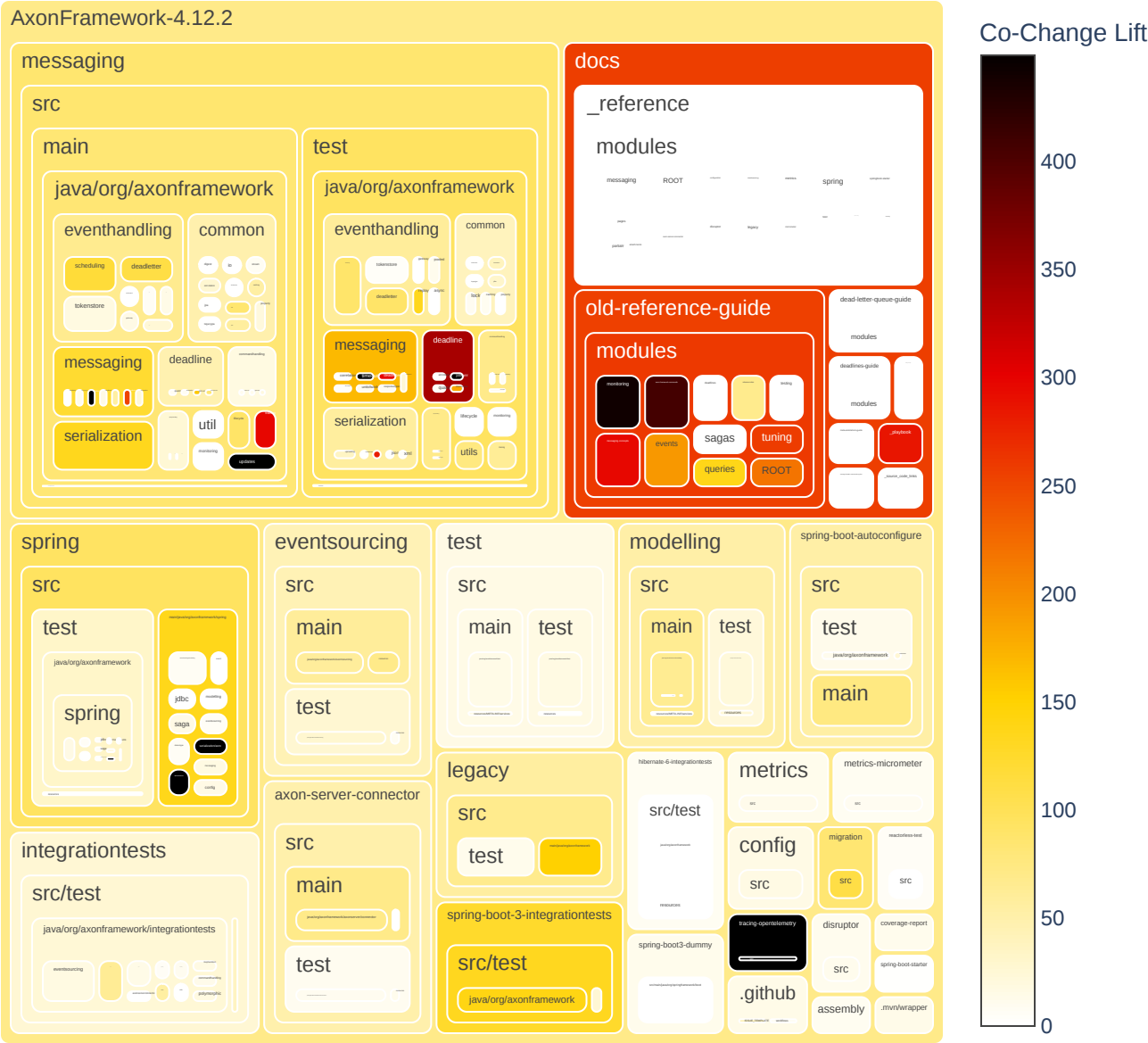
## Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

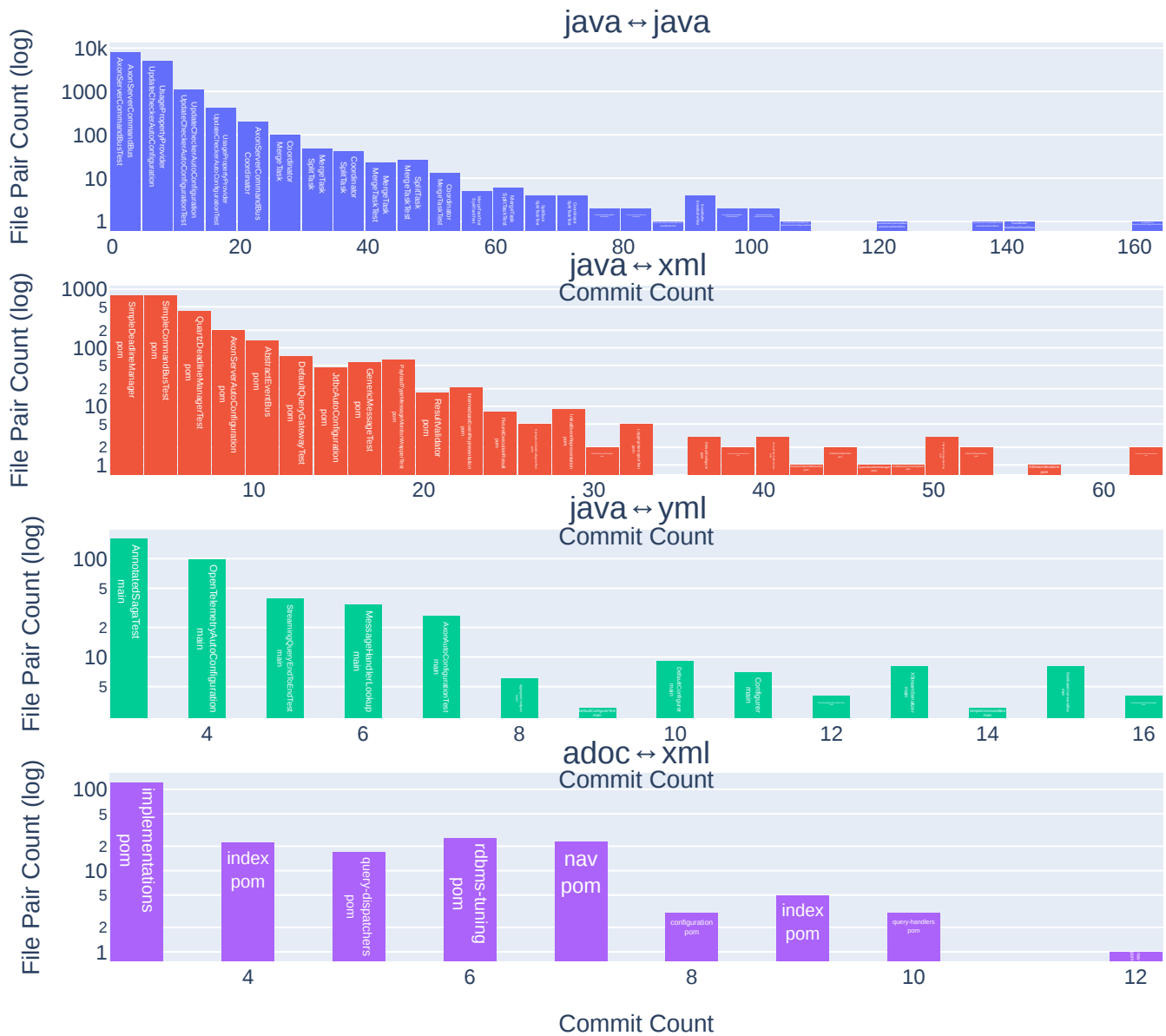


|   | fileExtensionPair | fileExtensionPairCount |
|---|-------------------|------------------------|
| 0 | java↔java         | 15199                  |
| 1 | java↔xml          | 2643                   |
| 2 | java↔yml          | 408                    |
| 3 | adoc↔xml          | 221                    |

Files changed together by commit count

|    | fileExtensionPair | updateCommitCount | GroupRank | filePair                                                                                     |
|----|-------------------|-------------------|-----------|----------------------------------------------------------------------------------------------|
| 0  | java↔java         | 163               | 1         | AxonServerCommandBus↔AxonServerQueryBus                                                      |
| 1  | java↔java         | 144               | 2         | AxonServerQueryBus↔AxonServerQueryBusTest                                                    |
| 2  | java↔java         | 135               | 3         | SpringAxonAutoConfigurerTest↔SpringAxonAutoConfigurer spring/src/tes                         |
| 3  | java↔java         | 121               | 4         | TrackingEventProcessorTest↔TrackingEventProcessor integrationtests/src/test/java/org/axonfra |
| 4  | java↔java         | 107               | 5         | SpringAxonAutoConfigurer↔AxonConfiguration spi                                               |
| 5  | java↔java         | 101               | 6         | AxonServerCommandBus↔AxonServerCommandBusTest                                                |
| 6  | java↔java         | 100               | 7         | EventProcessingModule↔EventProcessingModuleTest                                              |
| 7  | java↔java         | 96                | 8         | SagaTestFixture↔AggregateTestFixture                                                         |
| 8  | java↔java         | 95                | 9         | AxonServerCommandBusTest↔AxonServerQueryBusTest                                              |
| 9  | java↔java         | 94                | 10        | AggregateTestFixture↔FixtureConfiguration                                                    |
| 10 | java↔xml          | 62                | 1         | AggregateTestFixture↔pom                                                                     |
| 11 | java↔xml          | 57                | 2         | AggregateTestFixture↔pom                                                                     |
| 12 | java↔xml          | 53                | 3         | AggregateTestFixture↔pom                                                                     |
| 13 | java↔xml          | 51                | 4         | SpringAxonAutoConfigurer↔pom                                                                 |
| 14 | java↔xml          | 48                | 5         | SagaTestFixture↔pom                                                                          |
| 15 | java↔xml          | 46                | 6         | SpringAxonAutoConfigurer↔pom                                                                 |
| 16 | java↔xml          | 45                | 7         | SpringAxonAutoConfigurerTest↔pom                                                             |
| 17 | java↔xml          | 44                | 8         | SpringAxonAutoConfigurerTest↔pom                                                             |
| 18 | java↔xml          | 42                | 9         | AggregateTestFixture↔pom                                                                     |
| 19 | java↔xml          | 41                | 10        | AxonConfiguration↔pom                                                                        |
| 20 | java↔yaml         | 16                | 1         | JobRunrDeadlineManagerTest↔main                                                              |
| 21 | java↔yaml         | 15                | 2         | JobRunrEventSchedulerTest↔main                                                               |
| 22 | java↔yaml         | 14                | 3         | TrackingEventProcessor↔main                                                                  |
| 23 | java↔yaml         | 13                | 4         | SimpleQueryBus↔main                                                                          |
| 24 | java↔yaml         | 12                | 5         | SimpleEventHandlerInvokerTest↔main                                                           |
| 25 | java↔yaml         | 11                | 6         | AxonServerEventStore↔main                                                                    |
| 26 | java↔yaml         | 10                | 7         | PooledStreamingEventProcessor↔main                                                           |
| 27 | java↔yaml         | 9                 | 8         | AxonServerQueryBusTest↔main                                                                  |
| 28 | java↔yaml         | 8                 | 9         | DefaultConfigurer↔main                                                                       |
| 29 | java↔yaml         | 7                 | 10        | AggregateAnnotationCommandHandler↔main                                                       |
| 30 | adoc↔xml          | 12                | 1         | minor-releases↔pom                                                                           |
| 31 | adoc↔xml          | 10                | 2         | rdbms-tuning↔pom                                                                             |
| 32 | adoc↔xml          | 9                 | 3         | minor-releases↔pom                                                                           |
| 33 | adoc↔xml          | 8                 | 4         | minor-releases↔pom                                                                           |
| 34 | adoc↔xml          | 7                 | 5         | minor-releases↔pom                                                                           |
| 35 | adoc↔xml          | 6                 | 6         | nav↔pom                                                                                      |
| 36 | adoc↔xml          | 5                 | 7         | event-versioning↔pom                                                                         |
| 37 | adoc↔xml          | 4                 | 8         | rdbms-tuning↔pom                                                                             |
| 38 | adoc↔xml          | 3                 | 9         | implementations↔pom                                                                          |

## Co-Changed Files by their commit count

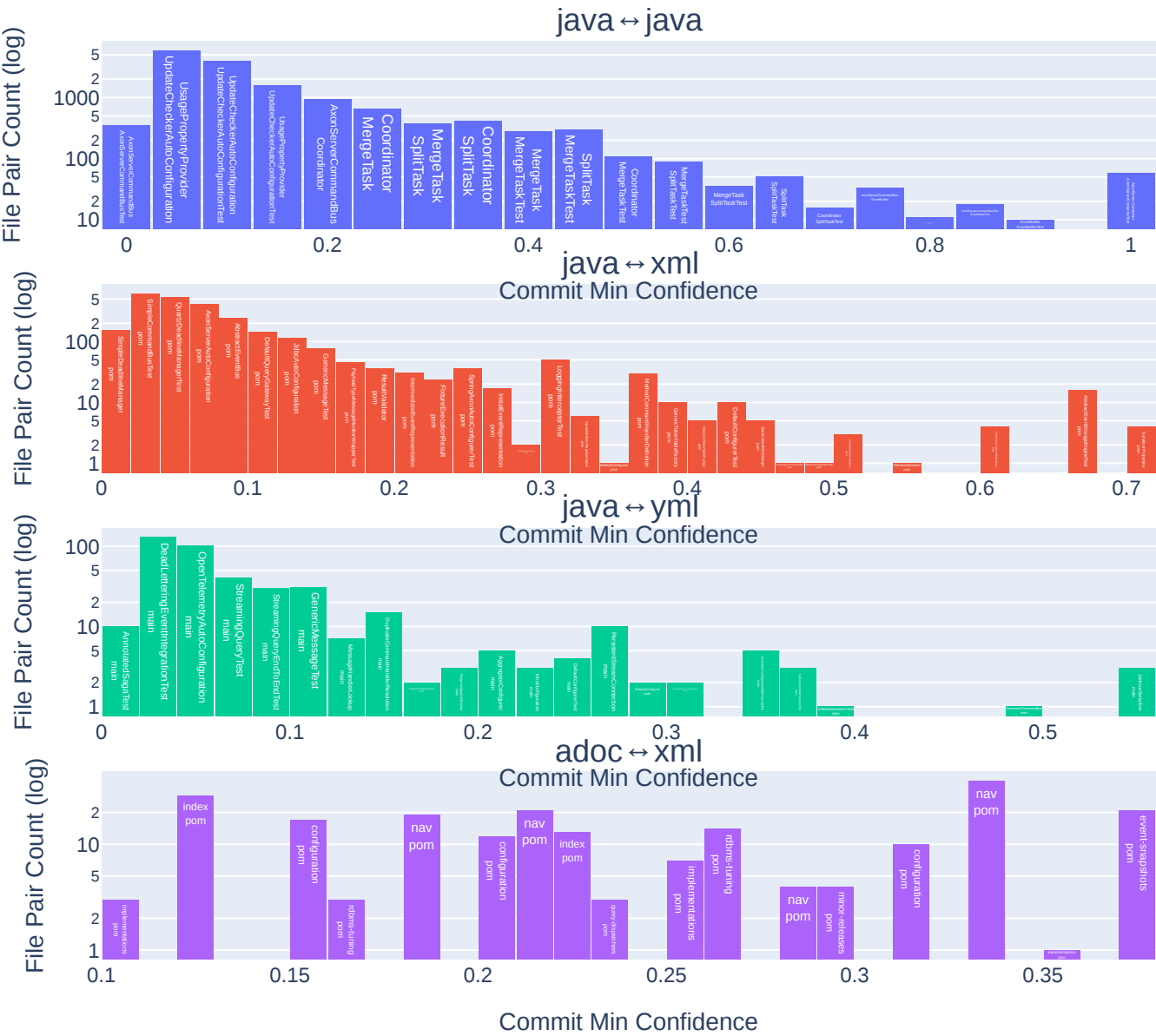


## Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

|    | fileExtensionPair | updateCommitMinConfidence | GroupRank | filePair                                                               |                |
|----|-------------------|---------------------------|-----------|------------------------------------------------------------------------|----------------|
| 0  | java↔java         | 1.000000                  | 1         | UsagePropertyProvider↔UpdateCheckerAutoConfiguration                   | mes            |
| 1  | java↔java         | 0.909091                  | 2         | AvroUtil↔SpecificRecordBaseSerializerStrategy                          |                |
| 2  | java↔java         | 0.900000                  | 3         | UnitOfWorkTimeoutInterceptor↔TimeoutWrappedMessageHandlingMember       |                |
| 3  | java↔java         | 0.875000                  | 4         | Decisions↔DoNotEnqueue                                                 |                |
| 4  | java↔java         | 0.857143                  | 5         | AvroSerializer↔AvroSerializerStrategy                                  |                |
| 5  | java↔java         | 0.833333                  | 6         | AvroSerializerStrategy↔ByteArrayToGenericRecordConverter               |                |
| 6  | java↔java         | 0.800000                  | 7         | DeadLetteringEventIntegrationTest↔InMemoryDeadLetteringIntegrationTest | messaging,     |
| 7  | java↔java         | 0.785714                  | 8         | AvroSerializer↔AvroSerializerTest                                      |                |
| 8  | java↔java         | 0.777778                  | 9         | JobrunrDeadlineManagerTest↔JobRunrEventSchedulerTest                   | integrationtes |
| 9  | java↔java         | 0.750000                  | 10        | AxonServerAutoConfiguration↔PersistentStreamMessageSource              |                |
| 10 | java↔xml          | 0.703704                  | 1         | JobRunrEventSchedulerTest↔pom                                          |                |
| 11 | java↔xml          | 0.666667                  | 2         | JobRunrEventSchedulerTest↔pom                                          |                |
| 12 | java↔xml          | 0.600000                  | 3         | ChainingConverter↔pom                                                  |                |
| 13 | java↔xml          | 0.545455                  | 4         | EventStoreTestUtils↔pom                                                |                |
| 14 | java↔xml          | 0.512821                  | 5         | DefaultEventMessageConverter↔pom                                       |                |
| 15 | java↔xml          | 0.500000                  | 6         | ChainingConverter↔pom                                                  |                |
| 16 | java↔xml          | 0.487179                  | 7         | DefaultEventMessageConverter↔pom                                       |                |
| 17 | java↔xml          | 0.461538                  | 8         | DefaultEventMessageConverter↔pom                                       |                |
| 18 | java↔xml          | 0.454545                  | 9         | JpaEventStoreAutoConfigurationWithoutAxonServerTest↔pom                |                |
| 19 | java↔xml          | 0.448276                  | 10        | JobrunrDeadlineManagerTest↔pom                                         |                |
| 20 | java↔yaml         | 0.555556                  | 1         | JobRunrEventSchedulerTest↔main                                         |                |
| 21 | java↔yaml         | 0.481481                  | 2         | JobRunrEventSchedulerTest↔antora                                       |                |
| 22 | java↔yaml         | 0.392857                  | 3         | AvroSerializerAutoConfiguration↔antora                                 |                |
| 23 | java↔yaml         | 0.375000                  | 4         | AxonTimeLimitedTaskTest↔main                                           |                |
| 24 | java↔yaml         | 0.357143                  | 5         | AvroSerializerAutoConfiguration↔main                                   |                |
| 25 | java↔yaml         | 0.340909                  | 6         | AbstractDeadlineManagerTestSuite↔main                                  |                |
| 26 | java↔yaml         | 0.300000                  | 7         | ChainingConverter↔main                                                 |                |
| 27 | java↔yaml         | 0.295455                  | 8         | AbstractDeadlineManagerTestSuite↔antora                                |                |
| 28 | java↔yaml         | 0.288889                  | 9         | JobrunrDeadlineManagerTest↔antora                                      |                |
| 29 | java↔yaml         | 0.277778                  | 10        | SpringAxonConfiguration↔main                                           |                |
| 30 | adoc↔xml          | 0.375000                  | 1         | index↔pom                                                              |                |
| 31 | adoc↔xml          | 0.352941                  | 2         | minor-releases↔pom                                                     |                |
| 32 | adoc↔xml          | 0.333333                  | 3         | nav↔pom                                                                |                |
| 33 | adoc↔xml          | 0.315789                  | 4         | query-dispatchers↔pom                                                  |                |
| 34 | adoc↔xml          | 0.312500                  | 5         | query-handlers↔pom                                                     |                |
| 35 | adoc↔xml          | 0.291667                  | 6         | event-snapshots↔pom                                                    |                |
| 36 | adoc↔xml          | 0.285714                  | 7         | implementations↔pom                                                    |                |
| 37 | adoc↔xml          | 0.281250                  | 8         | rbms-tuning↔pom                                                        |                |
| 38 | adoc↔xml          | 0.264706                  | 9         | minor-releases↔pom                                                     |                |
| 39 | adoc↔xml          | 0.263158                  | 10        | event-versioning↔pom                                                   |                |

# Co-Changed Files by their commit min confidence

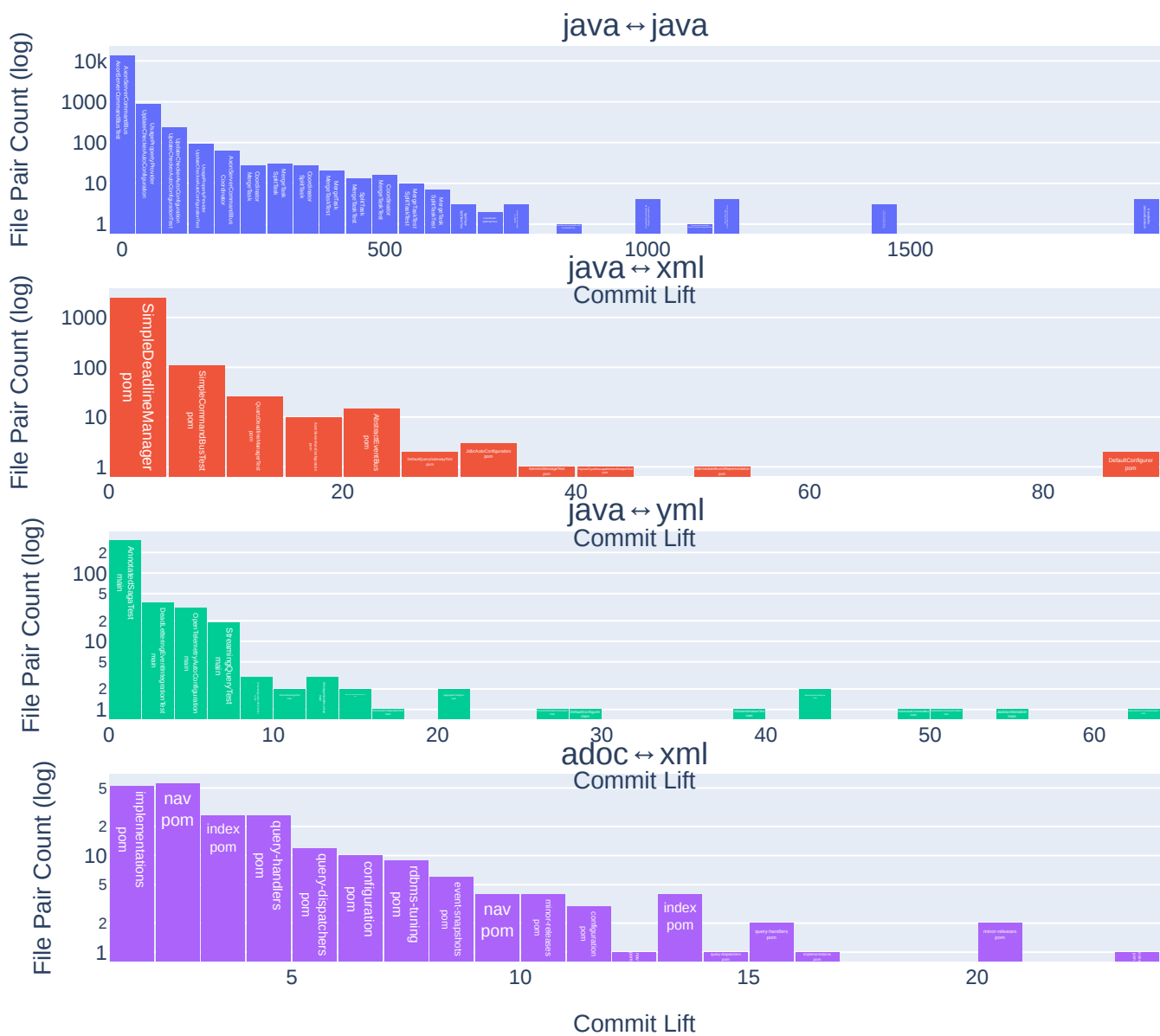


## Files changed together by commit lift



| fileExtensionPair | updateCommitLift | GroupRank   |    |
|-------------------|------------------|-------------|----|
| 0                 | java↔java        | 1950.333333 | 1  |
| 1                 | java↔java        | 1462.750000 | 2  |
| 2                 | java↔java        | 1170.200000 | 3  |
| 3                 | java↔java        | 1097.062500 | 4  |
| 4                 | java↔java        | 975.166667  | 5  |
| 5                 | java↔java        | 835.857143  | 6  |
| 6                 | java↔java        | 731.375000  | 7  |
| 7                 | java↔java        | 702.120000  | 8  |
| 8                 | java↔java        | 696.547619  | 9  |
| 9                 | java↔java        | 650.111111  | 10 |
| 10                | java↔xml         | 87.068452   | 1  |
| 11                | java↔xml         | 50.789931   | 2  |
| 12                | java↔xml         | 41.942652   | 3  |
| 13                | java↔xml         | 35.460606   | 4  |
| 14                | java↔xml         | 34.130833   | 5  |
| 15                | java↔xml         | 33.298374   | 6  |
| 16                | java↔xml         | 30.005128   | 7  |
| 17                | java↔xml         | 27.861905   | 8  |
| 18                | java↔xml         | 26.004444   | 9  |
| 19                | java↔xml         | 24.379167   | 10 |
| 20                | java↔yaml        | 62.603292   | 1  |
| 21                | java↔yaml        | 54.853125   | 2  |
| 22                | java↔yaml        | 51.080159   | 3  |
| 23                | java↔yaml        | 48.758333   | 4  |
| 24                | java↔yaml        | 43.926426   | 5  |
| 25                | java↔yaml        | 43.882500   | 6  |
| 26                | java↔yaml        | 38.415657   | 7  |
| 27                | java↔yaml        | 29.142912   | 8  |
| 28                | java↔yaml        | 26.954853   | 9  |
| 29                | java↔yaml        | 21.563882   | 10 |
| 30                | adoc↔xml         | 23.592742   | 1  |
| 31                | adoc↔xml         | 20.971326   | 2  |
| 32                | adoc↔xml         | 16.877885   | 3  |
| 33                | adoc↔xml         | 15.002564   | 4  |
| 34                | adoc↔xml         | 14.627500   | 5  |
| 35                | adoc↔xml         | 13.980884   | 6  |
| 36                | adoc↔xml         | 13.481567   | 7  |
| 37                | adoc↔xml         | 13.002222   | 8  |
| 38                | adoc↔xml         | 12.952878   | 9  |
| 39                | adoc↔xml         | 11.860135   | 10 |

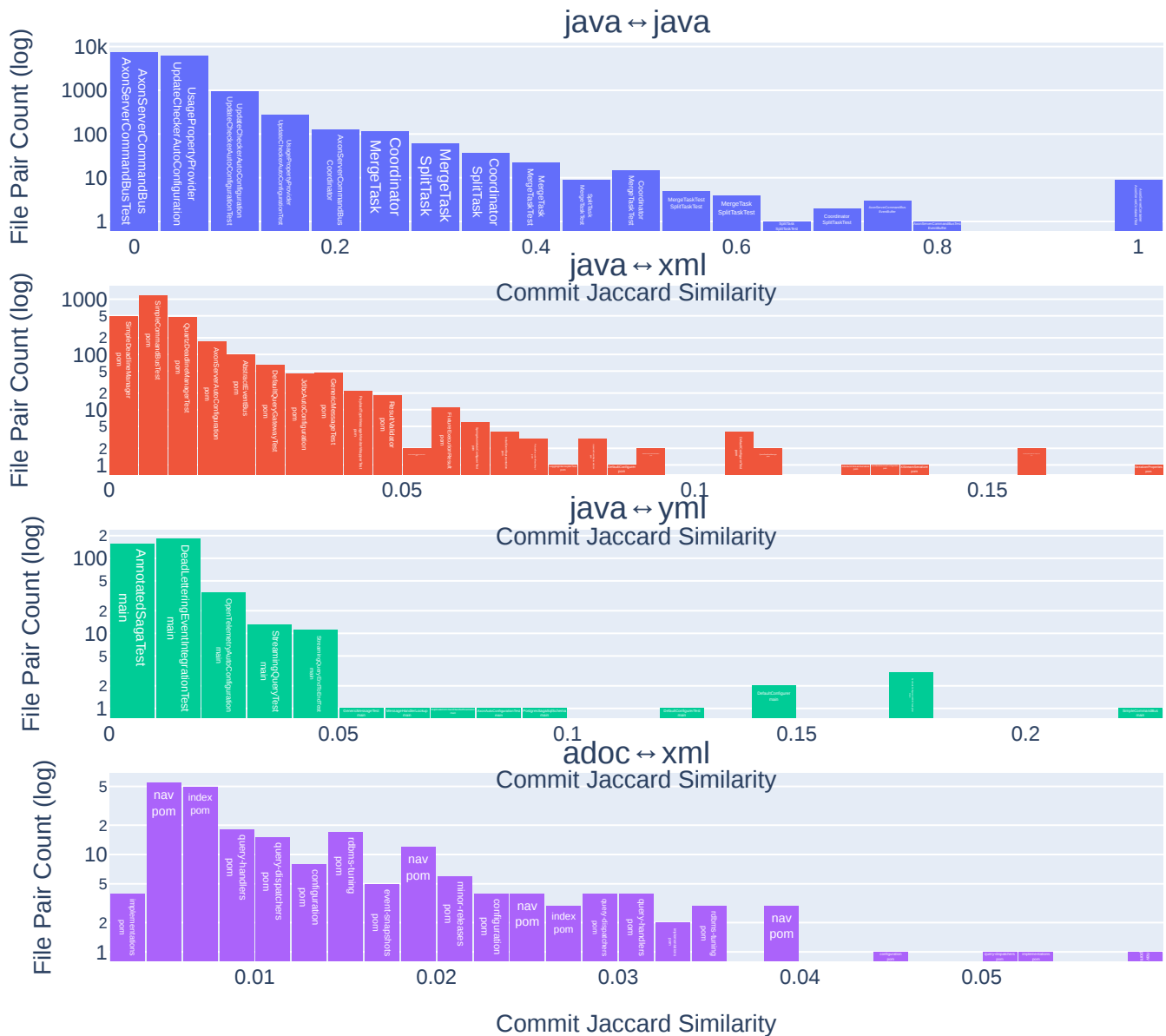
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

| file | ExtensionPair | updateCommitJaccardSimilarity | GroupRank | fileP                                                                               |
|------|---------------|-------------------------------|-----------|-------------------------------------------------------------------------------------|
| 0    | java↔java     | 1.000000                      | 1         | UsagePropertyProvider↔UpdateCheckerAutoConfigurat                                   |
| 1    | java↔java     | 0.800000                      | 2         | DoNotEnqueueTest↔ShouldEnqueueT                                                     |
| 2    | java↔java     | 0.750000                      | 3         | DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactory1                  |
| 3    | java↔java     | 0.714286                      | 4         | AvroUtil↔SpecificRecordBaseSerializerStrate                                         |
| 4    | java↔java     | 0.700000                      | 5         | DoNotEnqueue↔ShouldEnque                                                            |
| 5    | java↔java     | 0.625000                      | 6         | AvroSerializerStrategy↔ByteArrayToGenericRecordConver                               |
| 6    | java↔java     | 0.600000                      | 7         | UnitOfWorkTimeoutInterceptor↔TimeoutWrappedMessageHandlingMem                       |
| 7    | java↔java     | 0.555556                      | 8         | MessageAuthorizationDispatchInterceptorTest↔MessageAuthorizationHandlerInterceptorT |
| 8    | java↔java     | 0.545455                      | 9         | ShouldEnqueue↔DoNotEnqueueT                                                         |
| 9    | java↔java     | 0.529412                      | 10        | AnnotatedChildEntity↔ChildEn                                                        |
| 10   | java↔xml      | 0.176471                      | 1         | JobRunrEventSchedulerTest↔p                                                         |
| 11   | java↔xml      | 0.156250                      | 2         | TokenAddingInterceptor↔logbi                                                        |
| 12   | java↔xml      | 0.135338                      | 3         | JobrunrDeadlineManagerTest↔p                                                        |
| 13   | java↔xml      | 0.132231                      | 4         | AbstractDeadlineManagerTestSuite↔p                                                  |
| 14   | java↔xml      | 0.129496                      | 5         | JobRunrEventSchedulerTest↔p                                                         |
| 15   | java↔xml      | 0.113208                      | 6         | JobRunrEventSchedulerTest↔p                                                         |
| 16   | java↔xml      | 0.111111                      | 7         | TestAggregateRoot↔benchmark-mongo-cont                                              |
| 17   | java↔xml      | 0.109375                      | 8         | Message↔benchmark-mongo-cont                                                        |
| 18   | java↔xml      | 0.108280                      | 9         | AbstractDeadlineManagerTestSuite↔p                                                  |
| 19   | java↔xml      | 0.106383                      | 10        | UpstreamAwareStreamObserver↔logbi                                                   |
| 20   | java↔yaml     | 0.220339                      | 1         | JobRunrEventSchedulerTest↔ant                                                       |
| 21   | java↔yaml     | 0.177419                      | 2         | AvroSerializerAutoConfiguration↔ant                                                 |
| 22   | java↔yaml     | 0.174419                      | 3         | JobRunrEventSchedulerTest↔slack-release-notificat                                   |
| 23   | java↔yaml     | 0.171053                      | 4         | AbstractDeadlineManagerTestSuite↔ant                                                |
| 24   | java↔yaml     | 0.145631                      | 5         | AbstractDeadlineManagerTestSuite↔slack-release-notificat                            |
| 25   | java↔yaml     | 0.144444                      | 6         | JobrunrDeadlineManagerTest↔ant                                                      |
| 26   | java↔yaml     | 0.128205                      | 7         | JobrunrDeadlineManagerTest↔slack-release-notificat                                  |
| 27   | java↔yaml     | 0.090909                      | 8         | AbstractResourceInjector↔add-to-proj                                                |
| 28   | java↔yaml     | 0.081081                      | 9         | ConfigurationScopeAwareProvider↔add-to-proj                                         |
| 29   | java↔yaml     | 0.071429                      | 10        | AnnotatedMessageHandlingMemberTest↔ant                                              |
| 30   | adoc↔xml      | 0.058333                      | 1         | minor-releases↔p                                                                    |
| 31   | adoc↔xml      | 0.052632                      | 2         | nav↔p                                                                               |
| 32   | adoc↔xml      | 0.050420                      | 3         | rbms-tuning↔p                                                                       |
| 33   | adoc↔xml      | 0.044586                      | 4         | minor-releases↔p                                                                    |
| 34   | adoc↔xml      | 0.039735                      | 5         | nav↔p                                                                               |
| 35   | adoc↔xml      | 0.039548                      | 6         | minor-releases↔p                                                                    |
| 36   | adoc↔xml      | 0.038462                      | 7         | rbms-tuning↔p                                                                       |
| 37   | adoc↔xml      | 0.035971                      | 8         | rbms-tuning↔p                                                                       |
| 38   | adoc↔xml      | 0.035398                      | 9         | event-snapshots↔p                                                                   |
| 39   | adoc↔xml      | 0.035088                      | 10        | nav↔p                                                                               |

## Co-Changed Files by their commit jaccard similarity



## Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

|    | fileExtensionPair | filePair                                                                 | combinedMetricsScore | updateCommitCountExter |
|----|-------------------|--------------------------------------------------------------------------|----------------------|------------------------|
| 0  | adoc↔xml          | index↔pom                                                                | 24                   |                        |
| 1  | adoc↔xml          | nav↔pom                                                                  | 27                   |                        |
| 2  | adoc↔xml          | configuration↔pom                                                        | 28                   |                        |
| 3  | adoc↔xml          | nav↔pom                                                                  | 28                   |                        |
| 4  | adoc↔xml          | minor-releases↔pom                                                       | 30                   |                        |
| 5  | adoc↔xml          | rdbms-tuning↔pom                                                         | 36                   |                        |
| 6  | adoc↔xml          | nav↔pom                                                                  | 38                   |                        |
| 7  | adoc↔xml          | index↔pom                                                                | 39                   |                        |
| 8  | adoc↔xml          | rdbms-tuning↔pom                                                         | 41                   |                        |
| 9  | adoc↔xml          | minor-releases↔pom                                                       | 41                   |                        |
| 10 | java↔java         | UsagePropertyProvider↔UpdateCheckerAutoConfiguration                     | 84                   |                        |
| 11 | java↔java         | UpdateCheckerAutoConfiguration↔UpdateCheckerAutoConfigurationTest        | 84                   |                        |
| 12 | java↔java         | UsagePropertyProvider↔UpdateCheckerAutoConfigurationTest                 | 84                   |                        |
| 13 | java↔java         | AvroSchemaPackages↔AvroSchemaScan                                        | 84                   |                        |
| 14 | java↔java         | AvroSchemaPackages↔ClasspathAvroSchemaLoader                             | 84                   |                        |
| 15 | java↔java         | AvroSchemaScan↔ClasspathAvroSchemaLoader                                 | 84                   |                        |
| 16 | java↔java         | TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest | 84                   |                        |
| 17 | java↔java         | MetadataContextGetter↔MetadataContextSetter                              | 84                   |                        |
| 18 | java↔java         | DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest    | 87                   |                        |
| 19 | java↔java         | NoSuchDeadLetterException↔IgnoreTest                                     | 87                   |                        |
| 20 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 36                   |                        |
| 21 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 44                   |                        |
| 22 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 47                   |                        |
| 23 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 64                   |                        |
| 24 | java↔xml          | AbstractDeadlineManagerTestSuite↔pom                                     | 74                   |                        |
| 25 | java↔xml          | TokenAddingInterceptor↔logback                                           | 74                   |                        |
| 26 | java↔xml          | ContextAddingInterceptor↔logback                                         | 74                   |                        |
| 27 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 82                   |                        |
| 28 | java↔xml          | UpstreamAwareStreamObserver↔logback                                      | 83                   |                        |
| 29 | java↔xml          | EventStoreTestUtils↔persistence                                          | 85                   |                        |
| 30 | java↔yaml         | JobRunrEventSchedulerTest↔antora                                         | 8                    |                        |
| 31 | java↔yaml         | JobRunrEventSchedulerTest↔slack-release-notification                     | 11                   |                        |
| 32 | java↔yaml         | AvroSerializerAutoConfiguration↔antora                                   | 14                   |                        |
| 33 | java↔yaml         | AbstractDeadlineManagerTestSuite↔slack-release-notification              | 22                   |                        |
| 34 | java↔yaml         | AbstractDeadlineManagerTestSuite↔antora                                  | 23                   |                        |
| 35 | java↔yaml         | JobrunrDeadlineManagerTest↔antora                                        | 27                   |                        |
| 36 | java↔yaml         | AxonTimeLimitedTaskTest↔antora                                           | 33                   |                        |
| 37 | java↔yaml         | JobrunrDeadlineManagerTest↔slack-release-notification                    | 33                   |                        |
| 38 | java↔yaml         | JobRunrEventSchedulerTest↔pullrequest                                    | 34                   |                        |
| 39 | java↔yaml         | JobRunrEventSchedulerTest↔main                                           | 36                   |                        |

## Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

|    | filePair                                                                                              | combinedMetricsScore | updateCommitCount |
|----|-------------------------------------------------------------------------------------------------------|----------------------|-------------------|
| 0  | UsagePropertyProvider↔UpdateCheckerAutoConfiguration                                                  | 84                   |                   |
| 1  | UpdateCheckerAutoConfiguration↔UpdateCheckerAutoConfigurationTest                                     | 84                   |                   |
| 2  | UsagePropertyProvider↔UpdateCheckerAutoConfigurationTest                                              | 84                   |                   |
| 3  | AvroSchemaPackages↔AvroSchemaScan                                                                     | 84                   |                   |
| 4  | AvroSchemaPackages↔ClasspathAvroSchemaLoader                                                          | 84                   |                   |
| 5  | AvroSchemaScan↔ClasspathAvroSchemaLoader                                                              | 84                   |                   |
| 6  | AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor                                                      | 101                  | 1                 |
| 7  | TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest                              | 84                   |                   |
| 8  | AxonServerCommandBus↔AxonServerQueryBus                                                               | 3574                 | 16                |
| 9  | EventProcessingModule↔EventProcessingModuleTest                                                       | 2743                 | 10                |
| 10 | JobRunrDeadlineManagerTest↔JobRunrEventSchedulerTest                                                  | 331                  | 2                 |
| 11 | JobRunrEventSchedulerTest↔AbstractDeadlineManagerTestSuite                                            | 337                  | 1                 |
| 12 | SimpleEventHandlerInvoker↔SimpleEventHandlerInvokerTest                                               | 876                  | 4                 |
| 13 | DeadLetteringEventHandlerInvoker↔DeadLetteringEventHandlerInvokerTest                                 | 617                  | 2                 |
| 14 | JpaAutoConfiguration↔JpaJavaxAutoConfiguration                                                        | 725                  | 2                 |
| 15 | JpaAutoConfigurationTest↔JpaJavaxAutoConfigurationTest                                                | 298                  | 1                 |
| 16 | EventProcessingModule↔EventProcessingConfiguration                                                    | 2301                 | 3                 |
| 17 | EventPublicationOrderTest↔SynchronousLoopbackTest                                                     | 1557                 | 8                 |
| 18 | ResultValidatorImpl↔ResultValidator                                                                   | 1659                 | 9                 |
| 19 | FixtureExecutionResultImpl↔FixtureExecutionResult                                                     | 1301                 | 6                 |
| 20 | MetadataContextGetter↔MetadataContextSetter                                                           | 84                   |                   |
| 21 | ReflectionUtilsTest↔ReflectionUtils                                                                   | 935                  | 2                 |
| 22 | MultiSourceTrackingTokenTest↔MultiSourceTrackingToken                                                 | 939                  | 2                 |
| 23 | DoNotEnqueueTest↔ShouldEnqueueTest                                                                    | 93                   |                   |
| 24 | SpringAxonAutoConfigurer↔AxonConfiguration                                                            | 2850                 | 10                |
| 25 | AggregateMemberAnnotatedChildEntityCollectionDefinition↔AggregateMemberAnnotatedChildEntityDefinition | 287                  | 2                 |
| 26 | AggregateMemberAnnotatedChildEntityDefinition↔AggregateMemberAnnotatedChildEntityMapDefinition        | 290                  | 1                 |
| 27 | StubAggregate↔StubAggregate                                                                           | 826                  | 2                 |

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

|   | fileExtensionPair | filePair                                    | combinedMetricsScore | updateCommitCount | updateCommitMinConfidence | updateCommitRank |
|---|-------------------|---------------------------------------------|----------------------|-------------------|---------------------------|------------------|
| 0 | java↔java         | AggregateTestFixture↔TrackingEventProcessor | 14697                | 3                 | 0.007426                  |                  |

WordCloud of git authors

|   | word                | frequency |
|---|---------------------|-----------|
| 0 | Steven van Beelen   | 3552      |
| 1 | Allard Buijze       | 2741      |
| 2 | smcvb               | 822       |
| 3 | Milan Savic         | 449       |
| 4 | Mitchell Herrijgers | 327       |
| 5 | Gerard Klijs        | 246       |
| 6 | Marc Gathier        | 231       |
| 7 | sara pellegrini     | 163       |
| 8 | Rene de Waele       | 121       |
| 9 | Jettro Coenradie    | 117       |

Wordcloud of git authors

